LiveUtilities

A unified, persistent palette for real-time management of User Parameters, Configuration Snapshots, and Design Changelogs.

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Introduction: The "Why" and "What"

We've all been there:

- You are tweaking a design for a 3D print. You open `Modify > Change Parameters`. You change a value. You hit Enter. The modal box closes. You check the fit. It's wrong. You open the box again... **Lather. Rinse. Repeat.**
- You open a Fusion design you haven't touched in six months named `MyWidget_Final_v42`. You stare at the browser tree and wonder: *"Why did I add that chamfer? Is this actually the final version?"*

Fusion's native dialogs are functional, but they are often modal (blocking your view) and lack the space for saving iterative states or detailed historical context.

LiveUtilities solves this by combining three powerful tools—LiveParameters, LiveConfig, and Changelog Sidecar—into a single, modeless HTML palette that docks right inside Fusion. Instead of constantly opening and closing native dialogs, you have a persistent, tabbed interface to manage your design's math, states, and history in real-time.

Installation

Method: Manual Installation (Scripts & Add-Ins)

Currently, this add-in requires a quick manual installation.

1. Download the source code as a ZIP file and extract the `LiveUtilities` folder.
2. Move the entire `LiveUtilities` folder into your Fusion Add-Ins directory:
   - **Windows:** `%appdata%\Autodesk\Autodesk Fusion 360\API\AddIns`
   - **Mac:** `~/Library/Application Support/Autodesk/Autodesk Fusion 360/API/AddIns`
3. Open Fusion and press `Shift + S` to open the **Scripts and Add-Ins** dialog.
4. Click the **Add-Ins** tab at the top.
5. Click the green **"+"** icon next to "My Add-Ins" and select the `LiveUtilities` folder you just moved.
6. Select `LiveUtilities` from the list, check **"Run on Startup"** (optional), and click **Run**.
7. You will now find the **Live Utilities** button in the **Solid > Modify** panel.

Using LiveUtilities

The Global Interface

Clicking the Live Utilities button opens a persistent palette docked to the right side of your Fusion workspace.

- **Tabbed Navigation:** Seamlessly switch between Parameters, Config, and Changelog tools.
- **Theme Toggle:** Switch between Light and Dark mode using the toggle in the header to match your Fusion UI.
- **Auto-Sync & Global Refresh:** The palette automatically refreshes whenever you switch to a different active document. You can also manually hit the ↻ button in the header to rescan the model at any time.

Tab 1: Live Parameters

Keep your parameters docked on the side while you design. Tweak dimensions and see your model update instantly without closing windows.

- **Live Editing:** Type a new value or expression into any input box and press **Enter** (or click away) to apply it immediately.
- **Search & Filter:** Instantly filter your parameter list by name using the search bar, or toggle the ★ **Favs Only** switch to hide everything except your favorited parameters.
- **Creation:** Expand the "Add Parameter" section to create new ones on the fly. Supports Name, Unit (dropdown + custom), Expression, and Comments. *(Note: Text parameters must be enclosed in single quotes, e.g., `'MyText'`).*
- **Rename & Edit Comments:** Click the **Pencil (✎)** icon next to a parameter to safely rename it or update its comment.
- **Delete:** Click the **X** icon to remove a parameter. The add-in will prevent deletion if the parameter is currently in use by the model.

- **Safety Interlock:** To prevent data loss, edits are blocked while native Fusion commands (like Extrude, Fillet, or Sketch tools) are actively running. If you get an error, click the Fusion Canvas and press **ESC** to drop the active tool.

Tab 2: Live Config

The missing "Configuration Manager" for Fusion. Save specific combinations of parameters and feature states (Suppressed/Unsuppressed) as named Snapshots, acting as a "Poor Man's" configuration tool.

- **Snapshots:** Once you have your parameters and toggles set exactly how you like them, type a name (e.g., "Printer_A_Settings") and click **Save State**. Switch between snapshots with a single click.
- **Manage:** Use the 💾 **(Update)** button to overwrite a snapshot with the current screen state, or the 🗑️ **(Delete)** button to remove it.
- **Tracked Features (The `CFG_` Magic):** Want to toggle timeline features on and off?
    1. In the Fusion timeline, find a feature or group (Extrude, Fillet, Component Group).
    2. Rename it to start with `CFG_` (e.g., `CFG_Holes`).
    3. Click the Global Refresh (↻) button.
    4. You will now see a toggle switch for that feature in the palette, and its state will be saved in your Snapshots!
- **Data Locality:** Snapshots are stored as attributes *inside* the Fusion design file. If you share the file, the configurations travel with it.

Sample File: Download `Sink_Strainer_Live_Config_Demo.f3d` from the [Releases > Assets] section to test drive pre-configured snapshots and `CFG_` timeline groups.

Tab 3: Changelog Sidecar

A dedicated space to log your thoughts, decisions, and milestones. Because the logs are stored inside the Fusion file's attributes, the history travels with the design.

- **The Input Palette:**
    - **New Entry:** Type your notes here. Be verbose! Explain *why* you are making changes.
    - **Autosave Design:** Checked by default. Adding an entry will automatically save the Fusion design (creating a new version) to ensure the log is permanently attached. Uncheck to log a note for the current session without versioning immediately.
- **Utilities:**
    - **Create Milestone:** Reached a major turning point (e.g., "Prototype 1 Complete")? This archives the current active log into a history block and starts a fresh active log.
    - **Export:** Saves your entire history (Active + Milestones) to a `.txt` file on your computer.
- **The Sidecar Dashboard:** Click 📂 **OPEN LOG DASHBOARD** to launch a "Live View" of your history in your web browser.

- *Pro Tip:* Drag the browser tab out to create a separate floating window. Resize it into a narrow "Sidecar" next to your Fusion window or move it to a second monitor.
- *Auto-Refresh & Smart Scroll:* As you add entries in Fusion, the dashboard updates automatically and remembers your scroll position. Adjust the sync interval via the slider at the top of the dashboard.

Tech Stack

For the fellow coders and makers out there, here is how LiveUtilities was built:

- **Language:** Python (Fusion API)
- **Interface:** HTML5 / CSS3 / Vanilla JavaScript (running in a Fusion Palette)
- **Data Storage:** Custom JSON payloads stored in `Design.attributes` on the Root Component of the active design.
- **Communication:** Asynchronous JSON payload routing via `adsk.core.HTMLEventHandler`.
- **Dashboard Engine:** A custom generator that writes a localized, self-refreshing HTML file to the user's temporary directory, bypassing standard browser security restrictions for a seamless local experience.

Acknowledgements & Credits

- **Developer:** Ed Johnson (Making With An EdJ)
- **AI Assistance:** Developed with coding assistance from Google's Gemini 3.1 Pro model.
- **Icons:** "Lucy in the Sidecar" artwork generated via Artistly and enhamced with Gemini Nano Banana.
- **Lucy (The Cavachon Puppy):** ***Chief Wellness Officer & Director of Mandatory Breaks***
    - Thank you for ensuring I maintained healthy circulation and preventing Repetitive Strain Injury one fetch session at a time by interrupting my deep coding sessions.

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Happy Making! — EdJ